

Solution of Problem 1
By repeated application of

$$\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b)$$

and

$$\sin(a+b) = \sin(a)\cos(b) + \cos(a)\sin(b)$$

we find that

$$\cos^3(x) - \frac{3}{4}\cos(x) - \frac{1}{4}\cos(3x) = 0.$$

The real root in $\cos(x)$ of the above equation is

$$\cos(x) = \sqrt[3]{\frac{\frac{\cos(3x)}{4} - \sqrt{\frac{\cos^2(3x)}{16} - \frac{4 \cdot 27}{64 \cdot 27}}}{2}} + \sqrt[3]{\frac{\frac{\cos(3x)}{4} + \sqrt{\frac{\cos^2(3x)}{16} - \frac{4 \cdot 27}{64 \cdot 27}}}{2}},$$

which simplifies to:

$$\cos(x) = \frac{1}{2} \left(\sqrt[3]{\cos(3x) - i \sin(3x)} + \sqrt[3]{\cos(3x) + i \sin(3x)} \right). \quad (1)$$

Eq. (1) can be further simplified to

$$\cos(x) = \cos\left(\frac{1}{3} \arctan(\tan(3x))\right)$$

From Eq. (1) we also observe the following generalisation:

$$\cos(x) = \frac{1}{2} \left(\sqrt[m]{\cos(mx) - i \sin(mx)} + \sqrt[m]{\cos(mx) + i \sin(mx)} \right).$$